BUSINESS REQUIREMENTS DOCUMENT

EEG-Based Clinical Decision Support Tool

Product Feasibility & Business Analysis Study

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SCOPE NOTE: This BRD defines requirements for a proof-of-concept feasibility study. Requirements reflect what a real-world deployment would demand. The current implementation is analytical and exploratory and not clinically deployable in its present form.

1. Executive Summary

This Business Requirements Document defines the business and technical requirements for evaluating the feasibility of an AI-assisted clinical decision support system for schizophrenia risk classification using EEG-derived neurophysiological data.

The study assessed whether machine learning applied to NREM sleep biomarkers could support early-stage psychiatric screening workflows in a clinically responsible manner. The feasibility assessment concluded with a conditional go recommendation — acknowledging promising technical results while identifying critical gaps in sample size, external validation, and regulatory readiness that must be resolved before clinical deployment.

All requirements defined herein reflect the analytical, clinical, and operational considerations that would govern the real-world deployment of such a system.

2. Business Background & Problem Statement

Schizophrenia is a chronic psychiatric disorder affecting approximately 1% of the global population. It is characterized by significant cognitive, behavioral, and neurophysiological dysfunction — yet it typically goes undiagnosed for years after early neurological changes begin. The current diagnostic pathway relies primarily on:

- Clinical interviews and behavioral observation
- Subjective symptom assessment using standardized scales (e.g., PANSS, BPRS)
- Longitudinal monitoring over months to years

This process introduces diagnostic variability, delays intervention during the critical early window, and places high cognitive load on clinicians without objective decision-support tools.

Research has established that measurable neurophysiological changes — particularly in NREM sleep EEG spindle activity, slow oscillations, and spectral power — are detectable in schizophrenia patients and can be differentiated from healthy controls. These biomarkers represent a potential objective foundation for AI-assisted clinical decision support, enabling earlier and more structured psychiatric risk evaluation.

3. Business Objectives

1. Determine whether EEG-derived ML classification can achieve clinically meaningful accuracy (AUC ≥ 0.80) for schizophrenia risk stratification
2. Evaluate the operational feasibility of integrating such a system into existing psychiatric clinical workflows
3. Define the regulatory, ethical, and deployment requirements for a SaMD-class AI tool in psychiatric settings
4. Produce structured BA artifacts demonstrating the full product development lifecycle from concept to feasibility decision
5. Establish a go/no-go framework for future investment in EEG-based psychiatric decision support technology

4. Proposed Solution

The proposed solution is a proof-of-concept AI-assisted clinical decision support framework designed to:

- Process EEG-derived NREM sleep neurophysiological features
- Apply machine learning classification models to generate schizophrenia risk scores
- Produce structured, interpretable decision-support outputs for clinician review
- Integrate within existing psychiatric evaluation workflows as a supplementary analytical layer

The system is intended as a decision-support aid only. It does not replace clinical judgment, issue autonomous diagnoses, or recommend treatment. Final clinical decisions remain entirely with the treating psychiatrist.

5. Stakeholder Analysis

Stakeholder	Type	Role	Key Requirement
Psychiatrists / Clinicians	Primary	End Users	Accurate, explainable risk scores integrated into clinical workflow
Patients	Indirect	Beneficiaries	Safe, unbiased, privacy-protected system with human oversight
Hospital / Clinic Administration	Internal	Deployment Decision-Makers	Cost-effectiveness, regulatory compliance, operational fit
Healthcare IT Teams	Internal	Integration Owners	EHR compatibility, data security, and infrastructure requirements
Regulatory Bodies (FDA)	External	Compliance Authority	SaMD classification adherence, clinical validation evidence
Data Scientists / ML Engineers	Internal	Technical Builders	Feature pipeline stability, model versioning, performance monitoring
Compliance Officers	Internal	HIPAA & Ethics Oversight	Data anonymization, audit trails, consent frameworks

6. Current State — As-Is Process

The current psychiatric evaluation process for schizophrenia diagnosis follows this workflow:

1. Patient presents with behavioral or cognitive symptoms
2. Referral to a psychiatrist for a clinical interview
3. Structured symptom assessment using clinical scales (DSM-5 criteria)
4. Longitudinal observation over weeks to months
5. Diagnosis finalized based on clinical judgment
6. Treatment initiation — pharmacological and/or psychotherapy

Key Gaps in the As-Is Process

- No objective neurophysiological measurement at any stage of evaluation
- Diagnosis is entirely dependent on the clinician's interpretation and the patient's self-report
- Long diagnostic timelines delay early intervention during the critical risk window
- No structured mechanism to flag at-risk patients before full symptom onset
- Significant variability in diagnostic outcomes across clinical settings and practitioners

7. Future State — To-Be Process

The proposed AI-assisted workflow introduces EEG-based decision support as a structured supplementary layer within the existing psychiatric evaluation process:

1. Patients present with early behavioral or cognitive concerns
2. Standard clinical interview and symptom assessment conducted
3. EEG data acquisition during NREM sleep stage (overnight or abbreviated protocol)
4. Automated feature extraction from EEG signals across spindle, SO, spectral, and ERP domains
5. ML model generates risk classification score and probability output
6. Clinician reviews AI-generated report alongside clinical findings
7. Decision support output informs — but does not replace — clinical judgment
8. Diagnosis and treatment plan finalized by the psychiatrist

Key Improvements in the To-Be Process

- Objective neurophysiological data supplements subjective clinical assessment
- Earlier risk stratification is possible before full symptom onset
- Reduced diagnostic variability through standardized biomarker evaluation
- Structured decision support audit trail for clinical documentation
- Scalable across diverse clinical settings with consistent analytical methodology

8. Functional Requirements

FR-01	Data Ingestion

The system shall accept EEG-derived feature datasets in a structured tabular format. Input data must include NREM sleep neurophysiological metrics across spindles, slow oscillation, spectral power, coupling, and wake ERP domains.

- Supported format: CSV or equivalent structured tabular input
- Minimum required fields: spindle density, SO parameters, spectral power bands, coupling metrics
- Dataset used in feasibility study: 130 subjects, 2,894 numeric features

FR-02	Data Preprocessing

The system shall perform automated preprocessing before model inference, applied exclusively on training data to prevent data leakage.

- Missing value imputation: median strategy for numeric features, most-frequent for categorical
- Categorical encoding: One-Hot Encoding for demographic variables (AGE, SEX)
- Rare category handling: categories below 1% frequency threshold consolidated
- Duplicate removal and column-level quality checks

FR-03	Feature Selection

The system shall apply statistically validated feature selection to reduce the high-dimensional input space to clinically relevant biomarker subsets.

- Method: ANOVA F-test via SelectKBest
- Minimum K=100 features retained based on statistical significance
- Feature selection fitted on training data only — applied to test data post-fit
- Selected features dominated by fast spindle density (DENS_15) across frontal and central channels

FR-04	Risk Classification

The system shall generate a binary risk classification output (schizophrenia risk: positive/negative) along with a continuous probability score for clinical interpretation.

- Models evaluated: Logistic Regression, Decision Tree, Random Forest, SVM (RBF kernel)
- Best performing model: SVM — Accuracy 84.6%, AUC 0.896
- Minimum acceptable AUC threshold: 0.80
- Output: class label + probability score per patient record

FR-05	Model Evaluation Output

The system shall produce interpretable evaluation outputs for clinician and analyst review.

- ROC-AUC score and ROC curve visualization
- Confusion matrix with true/false positive and negative counts
- Precision, recall, F1-score, and classification report
- 5-fold stratified cross-validation results with mean and standard deviation

FR-06	Clinical Report Generation

The system shall generate a structured clinical summary report for clinician review.

- Risk classification result and confidence score
- Top contributing EEG biomarkers with feature importance scores
- Clear disclaimer: output is decision-support only, not a standalone diagnosis
- Patient record reference and timestamp for audit trail purposes

FR-07	Audit Trail

The system shall maintain a complete, timestamped audit log of all data inputs, preprocessing steps, model inferences, and report generations.

- Required for regulatory and compliance review
- Audit logs shall be tamper-evident and retained by applicable healthcare data governance standards
- All preprocessing transformations logged with parameters

FR-08	Workflow Integration

The system shall be designed for integration with existing EHR/EMR systems through standardized data interfaces.

- Interface standard: HL7 FHIR or equivalent clinical data exchange protocol
- Clinician access within existing workflow tools — no separate standalone application required
- Output report formatted for embedding within clinical documentation systems

9. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Model inference time is expected to remain within acceptable clinical workflow limits. Exact latency benchmarks would need to be defined during a formal system design phase based on deployment environment and EHR integration requirements.
NFR-02	Accuracy	Classification model shall maintain AUC ≥ 0.80 and F1-score ≥ 0.75 across diverse patient populations
NFR-03	Explainability	Model outputs shall include feature importance scores (ANOVA F-statistic-based) to support analytical interpretability. Full model explainability via SHAP or equivalent mechanism is identified as a mandatory pre-deployment requirement for any clinical implementation.
NFR-04	Security & Privacy	All patient data processed in compliance with HIPAA. Data transmission encrypted. No PII remained beyond clinical necessity

ID	Category	Requirement
NFR-05	Scalability	A deployable system would need to support multi-site clinical deployment without performance degradation. Cloud-based and on-premises deployment architectures should be evaluated during the system design phase as a deployment feasibility consideration.
NFR-06	Reliability	System reliability and uptime requirements would need to be defined in alignment with hospital IT infrastructure standards during the deployment planning phase. High availability is a recognized operational requirement for any clinical-facing system.
NFR-07	Regulatory Compliance	System developed in alignment with FDA SaMD guidance, including pre-market submission requirements per risk classification level
NFR-08	Interoperability	System shall support integration with major EHR platforms (Epic, Cerner) via standard clinical data interfaces

10. Data Requirements

Dimension	Detail
Input Features	EEG-derived NREM sleep metrics: spindle density/amplitude, slow oscillation parameters, spectral power bands (delta, theta, alpha, sigma, beta), spindle-SO coupling, wake auditory ERP
Feasibility Dataset	130 subjects (72 SCZ, 58 CTR), 2,894 numeric features, 3 categorical/demographic fields (DIS, AGE, SEX)
Missing Data Strategy	Median imputation for numeric features; most-frequent imputation for categorical fields; fitted on training set only
Feature Space Post-Selection	Top 100 features by ANOVA F-statistic; dominated by fast spindle density (DENS_15) across frontal and central channels
Deployment Data Source	Clinical EEG acquisition systems with an automated feature extraction pipeline
Privacy Requirement	Full de-identification required; IRB approval required for any clinical data collection
Recommended Dataset Size	Minimum 500+ subjects across diverse demographics for model retraining and clinical validation

11. Regulatory & Compliance Requirements

Requirement	Detail
FDA SaMD Classification	Likely Class II SaMD under FDA guidance, requiring 510(k) pre-market notification before deployment
HIPAA Compliance	Full compliance required for all patient data handling, storage, and transmission
IRB Approval	Required for any clinical data collection or human subjects research
Clinical Validation	Minimum one prospective clinical validation study required before deployment consideration
Bias & Fairness Audit	Required across age, sex, and demographic subgroups before regulatory submission
Post-Market Surveillance	Ongoing model performance monitoring requires post-deployment per FDA SaMD guidelines

12. Assumptions & Constraints

Assumptions

- EEG acquisition infrastructure exists at target clinical deployment sites
- Clinicians have baseline familiarity with neurophysiological monitoring concepts
- The ML model serves as a decision-support aid only — final diagnosis remains with the treating psychiatrist
- EEG-derived features contain clinically relevant neurophysiological patterns distinguishable by ML
- Any real deployment would require IRB approval and FDA SaMD clearance before patient use

Constraints

- Current feasibility study limited to 130 subjects — insufficient for clinical-grade validation
- Pre-extracted EEG features used; raw signal processing pipeline not implemented in this study
- No external validation dataset available for the current model
- High-dimensional feature space (~2,894 features) relative to sample size introduces overfitting risk
- Regulatory approval process reviewed conceptually only — formal submission not executed
- Model performance on demographically diverse populations has not yet been validated

13. Risk Assessment

Risk	Impact	Mitigation
Model overfitting due to a small sample size	High	5-fold stratified CV applied; results treated as exploratory; sample expansion recommended
False positives causing clinical misdiagnosis	High	System positioned as decision-support only; clinician oversight mandatory; threshold calibration required

Risk	Impact	Mitigation
False negatives cause missed early detection	High	Recalled metric monitored; conservative threshold recommended for clinical use; model retraining with larger data
Dataset bias across demographics	Medium	Age/sex distribution documented; diverse dataset required; bias audit required pre-deployment
Clinical misuse of AI-generated outputs	High	Explicit disclaimer on all outputs; training protocol for clinical users; governance policy required
Regulatory non-compliance	High	FDA SaMD framework reviewed; 510(k) pathway identified; formal submission required before deployment
Limited generalizability of the current model	High	External validation flagged as mandatory; Wave 2 replication by original authors noted as supporting evidence

14. Success Metrics

Success Criterion	Target	Achieved
ML model AUC on held-out test set	≥ 0.80	☑ SVM AUC: 0.896
Classification accuracy	≥ 80%	☑ SVM: 84.6%
Leakage-safe preprocessing pipeline	No data leakage	☑ Confirmed
Cross-validation stability	CV AUC ≥ 0.80	☑ RF: 0.845 ± 0.091
Interpretable evaluation metrics produced	All metrics present	☑ ROC, CM, F1, Report
Business requirements documented and traceable	BRD complete	☑ This document
Deployment risks and limitations identified	Documented	☑ Sections 12–13
Regulatory and responsible AI addressed	Documented	☑ Sections 11, 15

15. Feasibility Summary & Recommendation

Dimension	Assessment	Finding
Technical Feasibility	☑ Promising	SVM AUC 0.896, Accuracy 84% - promising but limited by sample size of 130 subjects
Clinical Feasibility	⚠ Conditional	Biomarkers validated in literature; clinical workflow integration viable in concept; validation study required

Dimension	Assessment	Finding
Regulatory Feasibility	✕ Not Ready	FDA SaMD pathway identified; 510(k) submission and clinical validation required before deployment
Operational Feasibility	⚠ Conditional	EHR integration and EEG infrastructure dependencies must be resolved at target clinical sites
Ethical Feasibility	⚠ Conditional	Bias auditing, an explainability framework, and responsible AI governance are required before deployment

Overall Recommendation: Conditional Go

The feasibility study demonstrated promising classification capability using EEG-derived neurophysiological features. The SVM model achieved an AUC of 0.896 and 84% classification accuracy on held-out test data, with cross-validation confirming reasonable model stability.

Continued development is recommended contingent on: expanded clinical dataset, external validation on independent cohorts, responsible AI governance implementation, FDA SaMD regulatory alignment, and HIPAA-compliant deployment infrastructure.

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